MediManage

Pharmacy Management and Medical Billing Desktop System

Research Documentation and Systems Architecture Report

Version 6.0.0

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Abstract and Scholarly Contribution

MediManage is a desktop-first pharmacy management system designed to improve the safety, accuracy, and operational efficiency of small to medium retail pharmacies. The project positions itself at the intersection of transactional integrity, user-centered workflow design, and intelligent decision support. Its computational core is a layered JavaFX application paired with an embedded SQLite store, ensuring offline resilience and predictable performance. Two adjacent microservices extend the core: a Python-based AI engine for care protocol generation and analytics, and a Node.js bridge for WhatsApp-based customer communication. The system emphasizes atomic billing, consistent stock governance, role-based access control, and extensible reporting pipelines. The expected impact is reduced operational error, faster checkout cycles, and stronger auditability, without requiring cloud dependence or specialized infrastructure.

Keywords: pharmacy management, transactional systems, JavaFX, SQLite, AI assistance, offline-first architecture

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1. Operational Context and Problem Formulation

Retail pharmacy operations are constrained by safety-critical inventory handling, high-frequency point-of-sale transactions, and regulatory audit requirements. Manual or semi-digital practices often fail to maintain accurate stock levels, leading to expired medicines, billing disputes, and weak traceability. Cloud-native solutions address some of these issues but introduce dependence on connectivity and centralized infrastructure. MediManage responds to this problem space with a local-first system that combines reliable transactional updates with an interface optimized for rapid data entry and minimal training overhead.

2. Research Objectives and Scope Delimitation

The project is guided by four objectives: (1) ensure transaction-safe billing and inventory updates; (2) unify customer, prescription, and procurement records into a consistent data model; (3) provide decision support through analytics and AI-derived care protocols; and (4) maintain operability in offline or low-connectivity settings. The scope includes billing, inventory, customers, prescriptions, suppliers, reporting, and messaging workflows. Excluded from the current scope are full multi-site synchronization, public marketplace integrations, and third-party payment gateways, which are positioned as future research extensions.

3. Methodological Foundation and Design Principles

MediManage adopts a layered MVC paradigm to separate presentation, orchestration, and persistence. The design enforces explicit boundaries between UI state, service-level business logic, and database access. This decoupling reduces defect propagation, supports incremental feature development, and creates a stable seam for future backend evolution. Architectural decisions are documented in formal ADRs, including a validated pathway for migration from pure SQLite to hybrid storage backends without requiring controller rewrites.

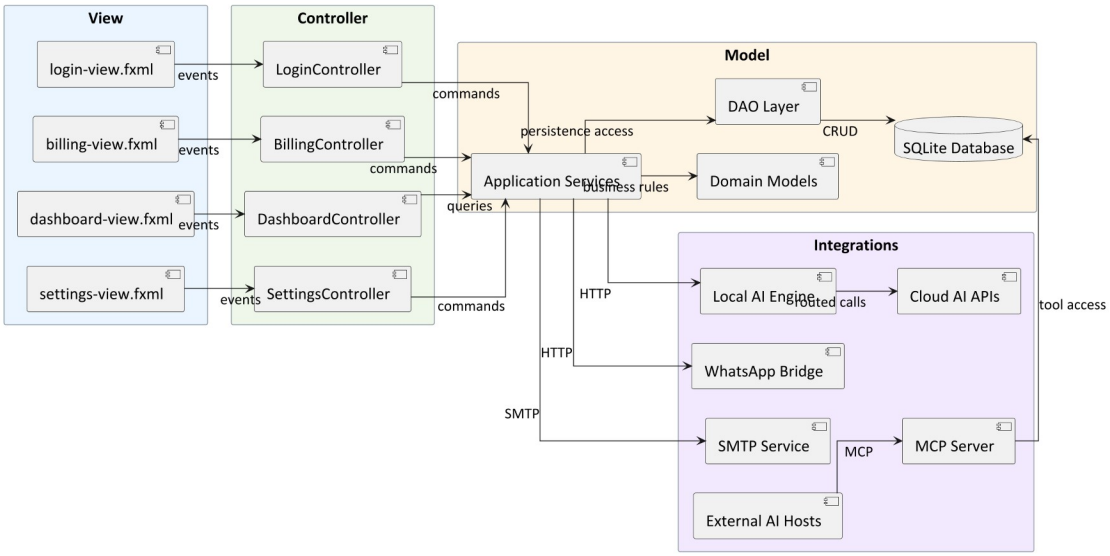

Figure 1. Layered MVC strategy with service boundaries

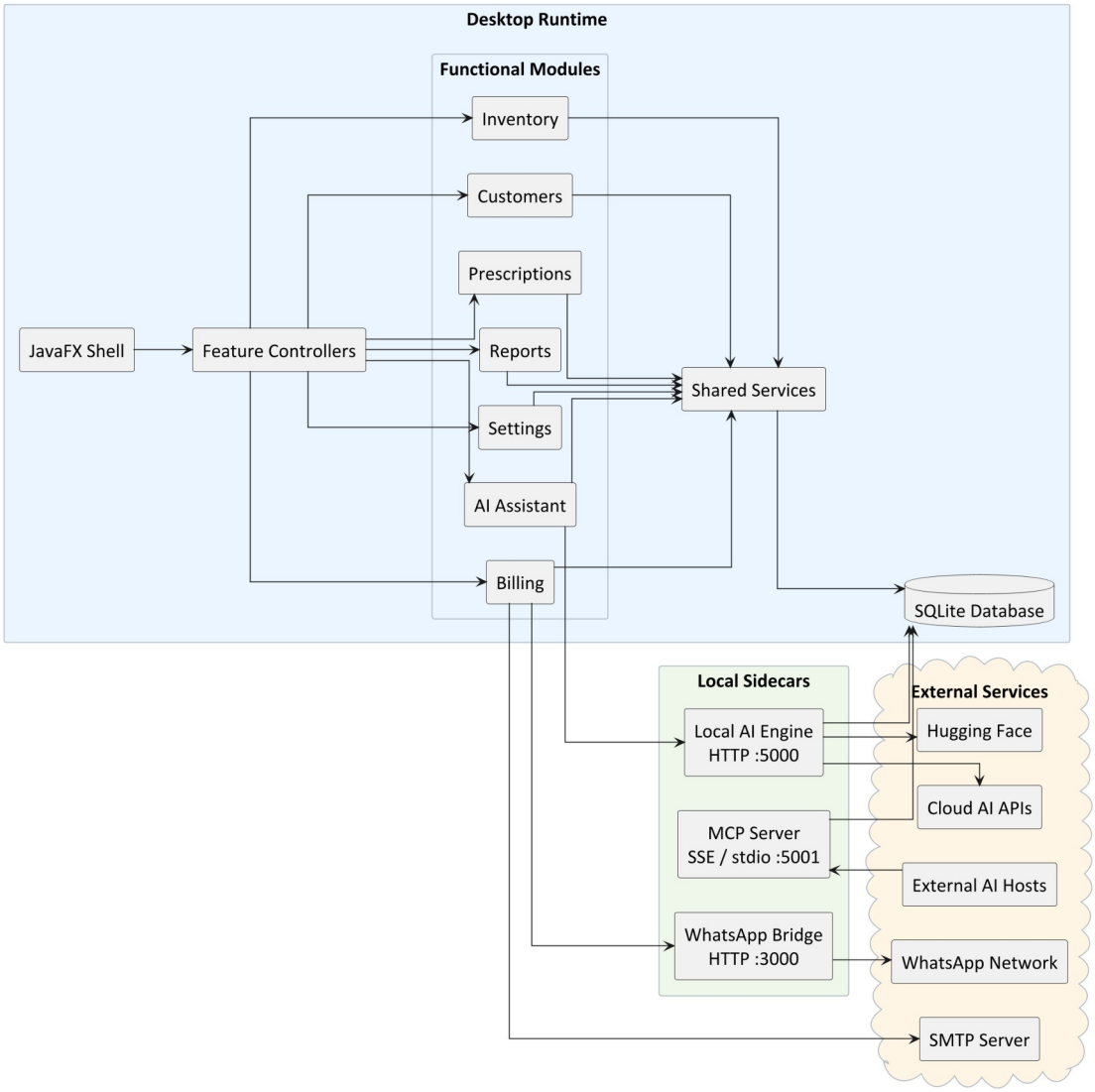

Figure 2. Module decomposition and subsystem boundaries

4. Subsystem Topology and Role Separation

The system is organized into four cooperative strata: the JavaFX presentation layer, an application service layer, a persistence layer composed of DAOs and schema migration logic, and external microservices. This topology reduces coupling, improves fault isolation, and enables optional components such as AI inference and WhatsApp dispatch to degrade gracefully without disrupting core operations.

Layer	Responsibilities
Presentation	User interaction, navigation, data binding, input validation
Application Services	Billing logic, reporting pipelines, inventory workflows, AI routing

Persistence	SQL access, schema migration, transactional integrity, indexing
External Integrations	AI microservice, WhatsApp bridge, SMTP delivery

5. Workflow Semantics and Transactional Guarantees

The billing pipeline is the system's most time-critical path. The workflow begins with medicine lookup by barcode or name, followed by cart aggregation, GST computation, and checkout. Checkout executes as a single ACID transaction that inserts bill headers, persists line items, updates stock quantities, records payment splits, and applies loyalty or credit adjustments. Failure in any step triggers rollback, thereby preventing inconsistent stock or orphaned invoices. This model is reinforced by schema constraints and runtime checks in DAO methods.

Inventory governance is maintained through explicit stock tables, adjustments for returns or damages, and expiry metadata. The schema supports reorder thresholds and optional multi-location extensions, enabling both immediate operational visibility and future scaling. Prescriptions, procurement, and attendance modules extend the system beyond point-of-sale into a broader pharmacy ERP context.

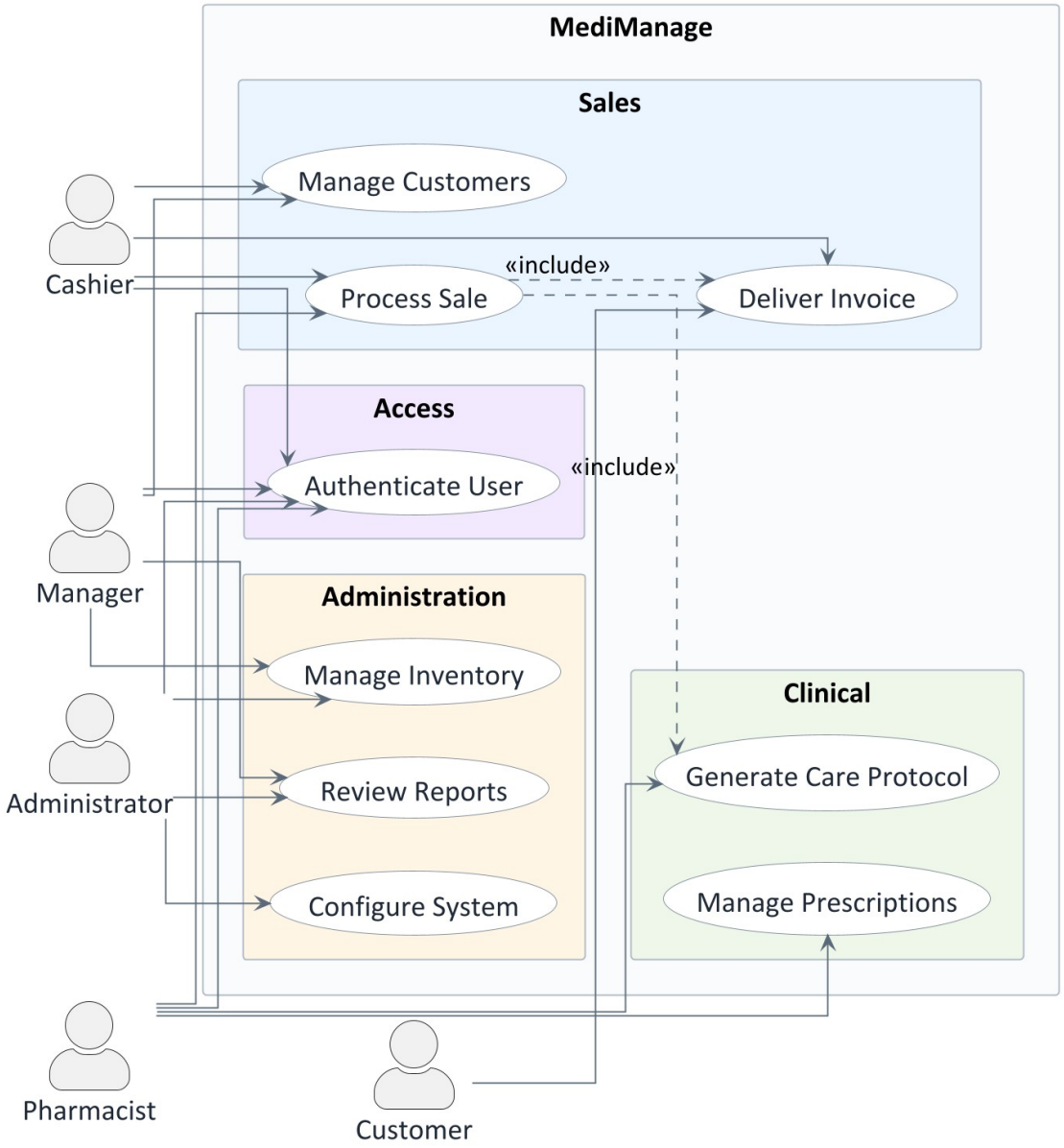

Figure 3. Use case coverage for pharmacy workflows

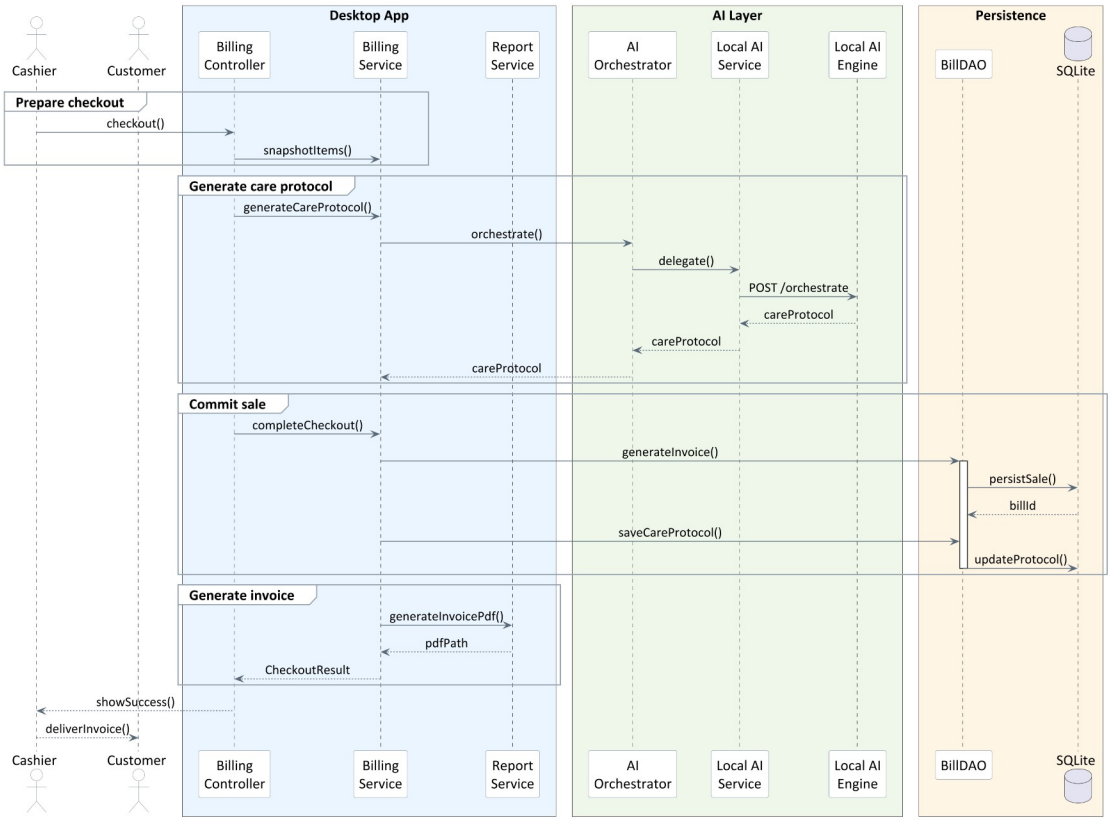

Figure 4. Billing and system interaction sequence

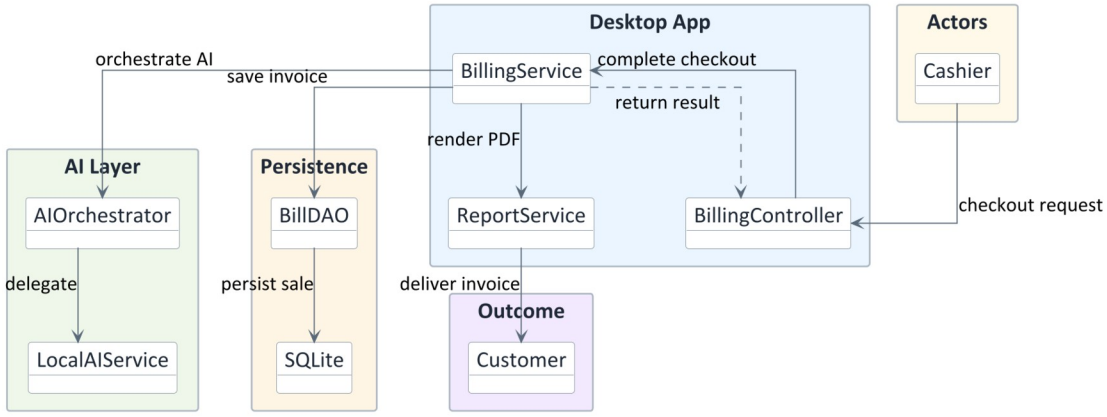

Figure 5. Collaboration among actors and services

6. Persistence Model, Schema Evolution, and Integrity Controls

SQLite is configured with WAL journaling and synchronous NORMAL mode to balance durability and throughput for local workloads. Schema initialization uses migration-safe logic that checks for existing columns before applying ALTER TABLE statements, minimizing risk when deploying to legacy environments. The schema is normalized around medicine, billing, customer, and supplier entities, with indexes optimized for

high-frequency queries such as name search, expiry detection, and temporal sales reporting.

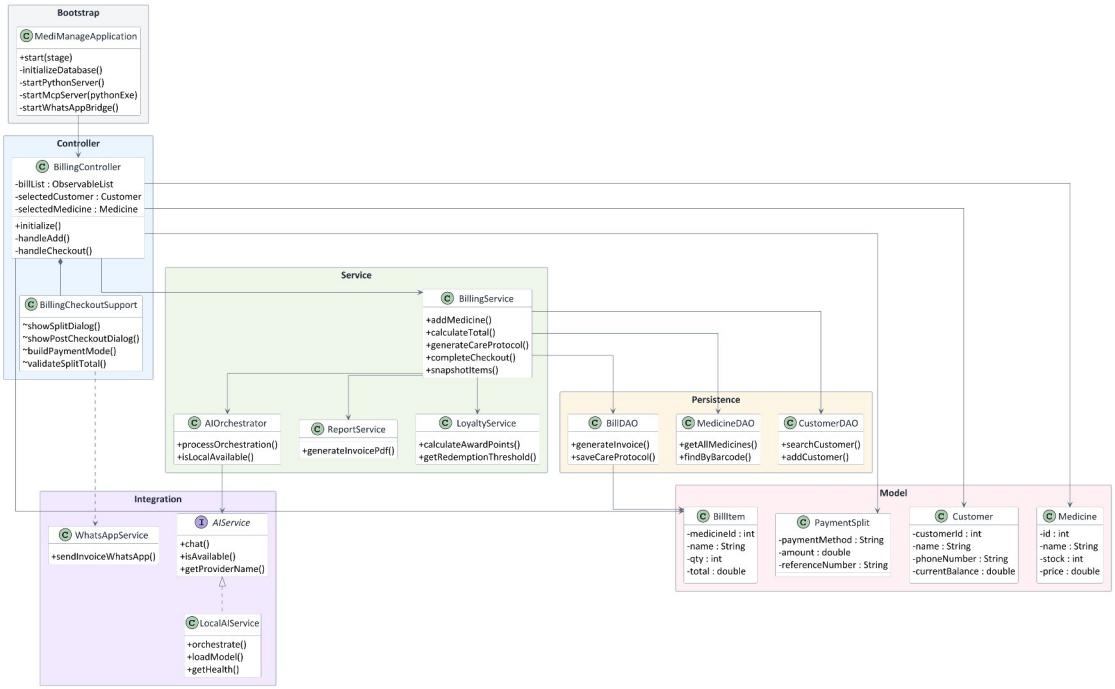

Figure 6. Core data model and entity relationships

7. Indexing Strategy and Query Optimization

Index design prioritizes query patterns critical to pharmacy operations: medicine name lookup, expiry filtering, daily sales ranges, and bill-item associations. The schema also includes composite indexes for anomaly tracking and report scheduling to support analytics dispatch workflows. These indexes directly reduce UI latency during high-volume interactions.

Index	Table	Columns
idx_users_username	users	username
idx_medicines_name	medicines	name
idx_medicines_active_name	medicines	active, name
idx_medicines_company	medicines	company
idx_medicines_expiry	medicines	expiry_date
idx_stock_medicine	stock	medicine_id
idx_bills_date	bills	bill_date
idx_bills_customer	bills	customer_id
idx_bill_items_bill	bill_items	bill_id
idx_bill_items_medicine	bill_items	medicine_id
idx_customers_phone	customers	phone

idx_expenses_date	expenses	date
idx_inventory_adjustments_medicine	inventory_adjustments	medicine_id
idx_inventory_adjustments_type_date	inventory_adjustments	adjustment_type, occurred_at
idx_anomaly_action_tracker_week	anomaly_action_tracker	week_start_date, timezone_name
idx_anomaly_action_tracker_status_due	anomaly_action_tracker	closure_status, due_date
idx_report_dispatch_schedule_next_run	analytics_report_dispatch_schedules	active, next_run_at
idx_report_dispatch_schedule_channel	analytics_report_dispatch_schedules	channel, recipient
idx_message_templates_key	message_templates	template_key
idx_medicines_generic	medicines	generic_name
idx_prescriptions_customer	prescriptions	customer_id
idx_prescriptions_status	prescriptions	status
idx_prescriptions_status_date	prescriptions	status, prescribed_date DESC
idx_ai_prompt_registry_active	ai_prompt_registry	prompt_key, is_active, version_number DESC
idx_ai_prompt_registry_history	ai_prompt_registry	prompt_key, version_number DESC

8. Data Dictionary and Entity Specifications

This section provides a structured data dictionary derived directly from the project schema. Each table is described in terms of columns, types, constraints, and inferred semantics. The dictionary is organized to support auditability, onboarding, and future schema evolution analysis.

8.1 users

Column	Type	Constraints	Description
user_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Identifier for application user (staff account)
username	TEXT	UNIQUE NOT NULL	Operational classification or state
password	TEXT	NOT NULL	Application data field
role	TEXT	DEFAULT 'STAFF'	Operational classification or

			state

8.2 medicines

Column	Type	Constraints	Description
medicine_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Identifier for medicine catalog entry
name	TEXT	NOT NULL	Application data field
generic_name	TEXT	-	Application data field
company	TEXT	-	Application data field
price	REAL	NOT NULL	Monetary value
expiry_date	TEXT	-	Timestamp or date value used for auditing or sequencing
active	INTEGER	DEFAULT 1	Boolean flag controlling behavior
barcode	TEXT	-	External identifier used for integration or compliance
supplier_id	INTEGER	-	System identifier or reference key
purchase_price	REAL	DEFAULT 0.0	Monetary value
reorder_threshold	INTEGER	DEFAULT 10	Application data field

8.3 stock

Column	Type	Constraints	Description
stock_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
medicine_id	INTEGER	UNIQUE	Foreign key to medicines.medicine_id
quantity	INTEGER	NOT NULL CHECK (quantity >= 0)	Quantity measure

8.4 customers

Column	Type	Constraints	Description
customer_id	INTEGER	PRIMARY KEY AUTOINCREMEN	Identifier for customer profile

		T	
name	TEXT	NOT NULL	Application data field
email	TEXT	-	Contact or communication field
phone	TEXT	-	Contact or communication field
address	TEXT	-	Contact or communication field
nominee_name	TEXT	-	Application data field
nominee_relation	TEXT	-	Application data field
insurance_provider	TEXT	-	Application data field
insurance_policy_no	TEXT	-	Application data field
diseases	TEXT	-	Application data field
photo_id_path	TEXT	-	Application data field
current_balance	REAL	DEFAULT 0.0	Application data field
loyalty_points	INTEGER	DEFAULT 0	Application data field

8.5 bills

Column	Type	Constraints	Description
bill_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
customer_id	INTEGER	-	Foreign key to customers.customer_id
user_id	INTEGER	-	Foreign key to users.user_id
total_amount	REAL	-	Monetary value
bill_date	TEXT	DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
payment_mode	TEXT	DEFAULT 'CASH'	Operational classification or

			state
ai_care_protocol	TEXT	-	AI related metadata
subscription_enrollment_id	INTEGER	-	System identifier or reference key
subscription_plan_id	INTEGER	-	System identifier or reference key
subscription_discount_percent	REAL	DEFAULT 0.0	Application data field
subscription_savings_amount	REAL	DEFAULT 0.0	Monetary value
subscription_approval_reference	TEXT	-	Application data field
location_id	INTEGER	-	System identifier or reference key

8.6 bill_items

Column	Type	Constraints	Description
item_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
bill_id	INTEGER	-	Foreign key to bills.bill_id
medicine_id	INTEGER	-	Foreign key to medicines.medicine_id
quantity	INTEGER	-	Quantity measure
price	REAL	-	Monetary value
total	REAL	-	Monetary value
subscription_discount_percent	REAL	DEFAULT 0.0	Application data field
subscription_discount_amount	REAL	DEFAULT 0.0	Monetary value
subscription_rule_source	TEXT	-	Application data field
price_override	REAL	-	Monetary value
override_reason	TEXT	-	Application data field
override_by_user_id	INTEGER	-	System identifier or reference key

8.7 expenses

Column	Type	Constraints	Description
expense_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
category	TEXT	NOT NULL	Application data

			field
amount	REAL	NOT NULL	Monetary value
date	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
description	TEXT	-	Free form text or narrative field

8.8 inventory_adjustments

Column	Type	Constraints	Description
adjustment_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
medicine_id	INTEGER	NOT NULL	Foreign key to medicines.medicine_id
adjustment_type	TEXT	NOT NULL	Application data field
quantity	INTEGER	NOT NULL CHECK (quantity > 0)	Quantity measure
unit_price	REAL	NOT NULL DEFAULT 0.0	Monetary value
root_cause_tag	TEXT	-	Application data field
notes	TEXT	-	Free form text or narrative field
occurred_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
created_by_user_id	INTEGER	-	Foreign key to users.user_id

8.9 anomaly_action_tracker

Column	Type	Constraints	Description
action_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
week_start_date	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
week_end_date	TEXT	NOT NULL	Timestamp or date value used for

			auditing or sequencing
timezone_name	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
alert_type	TEXT	NOT NULL	Application data field
severity	TEXT	NOT NULL	Application data field
metric_value	TEXT	-	Application data field
threshold_rule	TEXT	-	Application data field
alert_message	TEXT	-	Application data field
owner_user_id	INTEGER	-	Foreign key to users.user_id
due_date	TEXT	-	Timestamp or date value used for auditing or sequencing
closure_status	TEXT	NOT NULL DEFAULT 'OPEN'	Application data field
closed_at	TEXT	-	Timestamp or date value used for auditing or sequencing
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.10 analytics_report_dispatch_schedules

Column	Type	Constraints	Description
schedule_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
channel	TEXT	NOT NULL	Application data field
recipient	TEXT	NOT NULL	Application data field

report_format	TEXT	NOT NULL	Application data field
frequency	TEXT	NOT NULL	Application data field
timezone_name	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
filter_start_date	TEXT	-	Timestamp or date value used for auditing or sequencing
filter_end_date	TEXT	-	Timestamp or date value used for auditing or sequencing
supplier_filter	TEXT	-	Application data field
category_filter	TEXT	-	Application data field
active	INTEGER	NOT NULL DEFAULT 1	Boolean flag controlling behavior
next_run_at	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
last_run_at	TEXT	-	Timestamp or date value used for auditing or sequencing
last_status	TEXT	-	Application data field
last_error	TEXT	-	Application data field
created_by_user_id	INTEGER	-	Foreign key to users.user_id
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.11 message_templates

Column	Type	Constraints	Description
template_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
template_key	TEXT	UNIQUE NOT NULL	Application data field
subject	TEXT	-	Application data field
body_template	TEXT	NOT NULL	Free form text or narrative field
updated_at	TEXT	DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing

8.12 prescriptions

Column	Type	Constraints	Description
prescription_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
customer_id	INTEGER	-	Foreign key to customers.customer_id
customer_name	TEXT	NOT NULL	Application data field
doctor_name	TEXT	-	Application data field
status	TEXT	DEFAULT 'PENDING'	Operational classification or state
prescribed_date	TEXT	DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
notes	TEXT	-	Free form text or narrative field
medicines_text	TEXT	-	Application data field
ai_validation	TEXT	-	AI related metadata

8.13 ai_prompt_registry

Column	Type	Constraints	Description
prompt_version_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key

prompt_key	TEXT	NOT NULL	AI related metadata
version_number	INTEGER	NOT NULL	Application data field
template_text	TEXT	NOT NULL	Application data field
change_type	TEXT	NOT NULL	Application data field
change_note	TEXT	-	Application data field
rolled_back_from_version	INTEGER	-	Application data field
is_active	INTEGER	NOT NULL DEFAULT 0	Application data field
changed_by_user_id	INTEGER	-	Foreign key to users.user_id
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing

8.14 subscription_plans

Column	Type	Constraints	Description
plan_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
plan_code	TEXT	NOT NULL UNIQUE	Application data field
plan_name	TEXT	NOT NULL	Application data field
description	TEXT	-	Free form text or narrative field
price	REAL	NOT NULL DEFAULT 0.0	Monetary value
duration_days	INTEGER	NOT NULL DEFAULT 30	Application data field
grace_days	INTEGER	NOT NULL DEFAULT 0	Application data field
default_discount_percent	REAL	DEFAULT 0.0	Application data field
max_discount_percent	REAL	NOT NULL DEFAULT 0.0	Application data field
minimum_margin_percent	REAL	NOT NULL DEFAULT 0.0	Application data field
status	TEXT	NOT NULL DEFAULT 'ACTIVE'	Operational classification or state

auto_renew	INTEGER	NOT NULL DEFAULT 0	Boolean flag controlling behavior
requires_approval	INTEGER	NOT NULL DEFAULT 0	Boolean flag controlling behavior
created_by_user_id	INTEGER	-	Foreign key to users.user_id
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.15 payment_splits

Column	Type	Constraints	Description
split_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
bill_id	INTEGER	NOT NULL	Foreign key to bills.bill_id
payment_method	TEXT	NOT NULL	Application data field
amount	REAL	NOT NULL	Monetary value
reference_number	TEXT	-	External identifier used for integration or compliance
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.16 held_orders

Column	Type	Constraints	Description
hold_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
customer_id	INTEGER	-	Foreign key to customers.customer _id
user_id	INTEGER	-	Foreign key to users.user_id
items_json	TEXT	NOT NULL	Application data field
total_amount	REAL	NOT NULL DEFAULT 0.0	Monetary value
notes	TEXT	-	Free form text or

			narrative field
status	TEXT	NOT NULL DEFAULT 'HELD'	Operational classification or state
held_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing
recalled_at	TEXT	-	Timestamp or date value used for auditing or sequencing

8.17 receipt_settings

Column	Type	Constraints	Description
setting_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
pharmacy_name	TEXT	NOT NULL DEFAULT 'MediManage Pharmacy'	Application data field
address_line1	TEXT	-	Application data field
address_line2	TEXT	-	Application data field
phone	TEXT	-	Contact or communication field
email	TEXT	-	Contact or communication field
gst_number	TEXT	-	External identifier used for integration or compliance
logo_path	TEXT	-	Application data field
footer_text	TEXT	DEFAULT 'Thank you for your purchase!'	Application data field
show_barcode_on_ receipt	INTEGER	NOT NULL DEFAULT 1	Application data field
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.18 suppliers

Column	Type	Constraints	Description
supplier_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
name	TEXT	NOT NULL	Application data field
contact_person	TEXT	-	Application data field
phone	TEXT	-	Contact or communication field
email	TEXT	-	Contact or communication field
address	TEXT	-	Contact or communication field
gst_number	TEXT	-	External identifier used for integration or compliance
active	INTEGER	NOT NULL DEFAULT 1	Boolean flag controlling behavior
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing

8.19 purchase_orders

Column	Type	Constraints	Description
po_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
supplier_id	INTEGER	NOT NULL	Foreign key to suppliers.supplier_id
order_date	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
expected_delivery	TEXT	-	Application data

			field
status	TEXT	NOT NULL DEFAULT 'DRAFT'	Operational classification or state
total_amount	REAL	NOT NULL DEFAULT 0.0	Monetary value
notes	TEXT	-	Free form text or narrative field
created_by_user_id	INTEGER	-	Foreign key to users.user_id
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIME STAMP	Timestamp or date value used for auditing or sequencing

8.20 purchase_order_items

Column	Type	Constraints	Description
poi_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
po_id	INTEGER	NOT NULL	Foreign key to purchase_orders.po _id
medicine_id	INTEGER	NOT NULL	Foreign key to medicines.medicine _id
ordered_qty	INTEGER	NOT NULL	Application data field
received_qty	INTEGER	NOT NULL DEFAULT 0	Application data field
unit_cost	REAL	NOT NULL	Monetary value

8.21 employee_attendance

Column	Type	Constraints	Description
attendance_id	INTEGER	PRIMARY KEY AUTOINCREMEN T	System identifier or reference key
user_id	INTEGER	NOT NULL	Foreign key to users.user_id
date	TEXT	NOT NULL	Timestamp or date value used for auditing or sequencing
total_hours	REAL	-	Monetary value
notes	TEXT	-	Free form text or narrative field

8.22 locations

Column	Type	Constraints	Description
location_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
name	TEXT	NOT NULL	Application data field
address	TEXT	-	Contact or communication field
phone	TEXT	-	Contact or communication field
location_type	TEXT	NOT NULL DEFAULT 'PHARMACY'	Application data field
active	INTEGER	NOT NULL DEFAULT 1	Boolean flag controlling behavior
created_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing

8.23 location_stock

Column	Type	Constraints	Description
location_stock_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
location_id	INTEGER	NOT NULL	Foreign key to locations.location_id
medicine_id	INTEGER	NOT NULL	Foreign key to medicines.medicine_id
quantity	INTEGER	NOT NULL DEFAULT 0	Quantity measure
min_stock	INTEGER	NOT NULL DEFAULT 5	Application data field
updated_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing

8.24 stock_transfers

Column	Type	Constraints	Description

transfer_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System identifier or reference key
from_location_id	INTEGER	NOT NULL	Foreign key to locations.location_id
to_location_id	INTEGER	NOT NULL	Foreign key to locations.location_id
medicine_id	INTEGER	NOT NULL	Foreign key to medicines.medicine_id
quantity	INTEGER	NOT NULL	Quantity measure
status	TEXT	NOT NULL DEFAULT 'PENDING'	Operational classification or state
requested_by	INTEGER	-	Foreign key to users.user_id
requested_at	TEXT	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp or date value used for auditing or sequencing
completed_at	TEXT	-	Timestamp or date value used for auditing or sequencing

9. Security, Privacy, and Access Governance

MediManage employs role-based access control, with administrative permissions restricted to privileged roles. Passwords are hashed using BCrypt with a defined work factor, and legacy plaintext values are supported for migration compatibility. Sensitive secrets, including SMTP credentials and AI provider keys, are secured using Windows DPAPI through a local secure store. These mechanisms collectively reduce exposure risk while maintaining usability in offline environments.

10. Intelligence Augmentation and AI Orchestration

The AI subsystem operates as a Python microservice that exposes endpoints for model download, orchestration, and inference. The Java layer includes an orchestrator that enforces rate limiting, maintains an LRU cache, and supports routing between local and cloud providers. The AI pipeline is designed to degrade gracefully: a rules-based care protocol engine is invoked when AI responses are unavailable or unsafe. This maintains continuity in pharmacy guidance generation and supports ethical deployment constraints.

11. Communication and Interoperability Layer

MediManage integrates external communication through a Node.js WhatsApp bridge. The bridge manages QR-based authentication and provides endpoints for message and PDF delivery. In the desktop application, messages are templated using database-stored content, sanitized, and dispatched asynchronously to preserve UI responsiveness. A parallel SMTP pipeline enables invoice delivery via email.

12. Human Factors and Interface Ergonomics

The UI is designed for rapid task completion and low cognitive load. Controls are grouped by workflow stage, and JavaFX bindings provide reactive updates across tables, forms, and dashboards. The following placeholders indicate required visual evidence to accompany the narrative.

==[INSERT SCREENSHOT OF LOGIN SCREEN HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF DASHBOARD OVERVIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF BILLING AND CART VIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF INVENTORY MANAGEMENT VIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF CUSTOMER MANAGEMENT VIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF REPORTS AND ANALYTICS VIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF SETTINGS AND AI CONFIGURATION VIEW HERE]==(highlighed with colour and size)

==[INSERT SCREENSHOT OF INVOICE PDF OUTPUT HERE]==(highlighed with colour and size)

13. Verification Strategy and Test Coverage

The project includes unit and integration tests covering billing integrity, DAO operations, role policies, AI input safety, and storage concurrency. Maven Surefire is configured to fail builds when no tests are detected, preserving baseline quality gates. The test suite emphasizes high-risk paths such as invoice generation, loyalty adjustments, and stock decrement behavior under concurrent access.

14. Evaluation Methodology and Comparative Analysis

A formal evaluation framework is required to validate system benefits beyond anecdotal evidence. This report proposes a multi-dimensional evaluation protocol, grounded in pharmacy operations research and software quality metrics.

14.1 Proposed Metrics and Instrumentation

Key performance and reliability metrics include: checkout latency (time from cart finalization to invoice creation), stock accuracy (difference between system stock and physical count), billing error rate (number of corrected invoices per 100 transactions), and data completeness (percentage of bills with linked customer and payment metadata). Logging hooks and report exports can be used to collect these metrics without interrupting operations.

14.2 Comparative Baselines

Two baselines are recommended for comparison. The first is manual ledger or spreadsheet workflows, which typically exhibit higher latency and lower auditability. The second is cloud-first pharmacy software, which provides centralized analytics but is sensitive to connectivity and operational cost. MediManage is expected to outperform manual systems in accuracy and speed while matching or exceeding cloud systems in reliability under limited network conditions.

14.3 Reliability and Resilience Assessment

Reliability can be evaluated through controlled failure injection, including forced AI service downtime, messaging service unavailability, and simulated database contention. The system should continue core billing and inventory operations under these scenarios, with graceful degradation of auxiliary services.

14.4 Ethical and Safety Evaluation

AI-generated care protocols must be validated for safety and alignment with pharmacy guidelines. A review protocol should compare AI suggestions to standard care references

and document deviations. The fallback rules engine provides a baseline that can be used as a safety comparison benchmark.

15. Limitations, Risks, and Future Research Directions

Current deployment is optimized for single-site usage with optional in-memory concurrency testing. Multi-site synchronization, role-based audit logging, and formal regulatory compliance audits are potential expansions. The AI subsystem could be extended to drug interaction detection, personalized adherence reminders, and predictive inventory forecasting. Future work should also include usability studies, performance profiling under high transaction loads, and longitudinal impact assessment on billing error reduction.

16. Conclusion

MediManage represents a comprehensive, research-aligned pharmacy management platform that balances transactional correctness, human-centered workflow design, and intelligent augmentation. Its layered architecture, migration-aware data model, and robust billing pipeline provide a stable foundation for real-world adoption. The inclusion of AI and messaging services introduces differentiated value while preserving offline resilience. The system is positioned as a strong final-year project with clear pathways for future research and production enhancement.

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